

Bixi Project - Data Analysis in Tableau

Introduction

Bixi is a bike sharing and bike rental company established in Montreal, Quebec.¹ The BI Manager from Bixi as well as the marketing team and the Director of Finance ordered this report to further derive insights from data. This report is created to support the executive team at Bixi in making business decisions and growing their company. In this report, I examined and visualized the data from Bixi using visual analytics in Tableau. Additionally, I utilized information on Bixi's official website and Google Maps to supplement my findings.

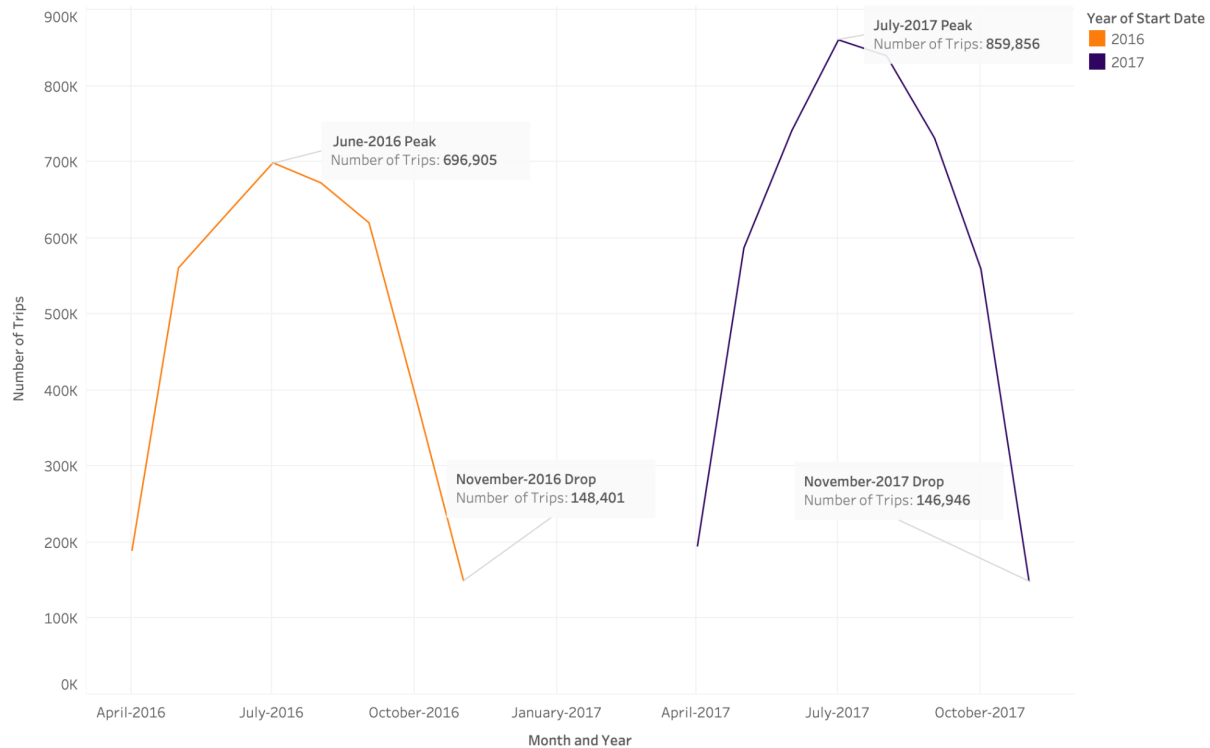
Bixi Use Growth from 2016 to 2017

I looked at the overall usage growth between 2016 and 2017 broken down by month. The visualization below demonstrates a clear increase in Bixi usage from 2016 to 2017. On a month-to-month basis, the usage of Bixi bikes was generally higher in 2017 compared to 2016 with only exception being November. For example, in July of 2017, there were a total of **859,856** trips, whereas in July of 2016, there were **696,905** trips. The graph below demonstrates the usage growth between the two years.

¹ *Who we are*. BIXI Montréal. (n.d.). Retrieved April 16, 2023, from <https://bixi.com/en/who-we-are>

The Number of Trips for 2016 - 2017 broken down by Month

The usage of Bixi bikes has increased in 2017 compared to 2016

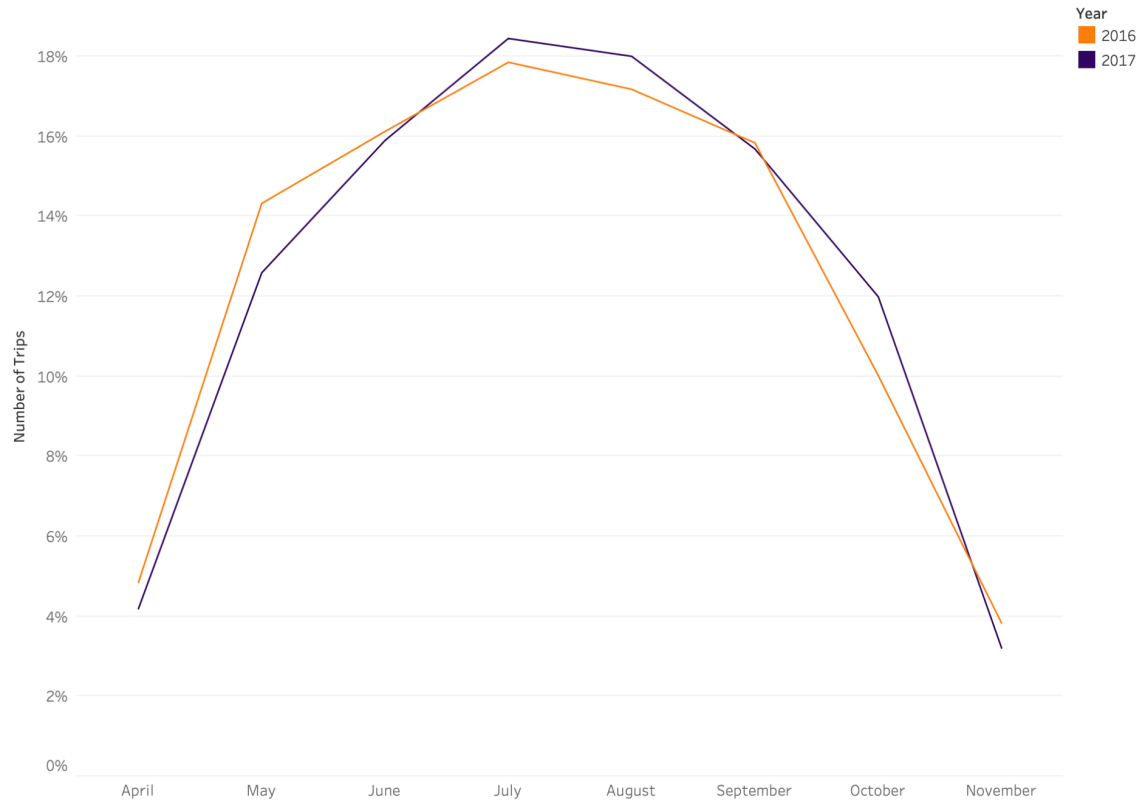


The trend of count of Id for Start Date Month. Color shows details about Start Date Year. The view is filtered on Start Date Year, which keeps 2016 and 2017.

The next graph looks at the percentage of Bixi trips per month in 2016 and 2017. It is to note that the percentages are not compared against each other. Rather the percentages per month are calculated out of the total percentage of trips in their respective years. The data on the graph revealed that both years saw the same pattern of usage. Specifically, the usage tended to be lower in early spring and late fall, but increased during summer months in both years.

Percentages of Trips per Month in 2016 and 2017

The percentages per month are calculated out of a respective year (2016, 2017)



The trend of % of Total Number of Trips for Start Date Month. Color shows details about Start Date Year.

Member and Non-Member Bixi Use

Members used Bixi bikes more often than Non-Members. The graph below illustrates that on average **Members** made around **80%** of all trips in 2017.

Percentage of Trips by Members in 2017

Members on average make around **80%** of total trips

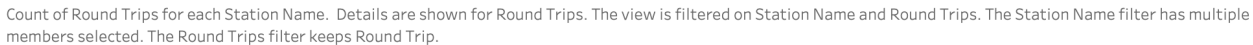


The trend of Percentage trips members for Start Date Month. The marks are labeled by Percentage trips members. The data is filtered on Start Date Year, which keeps 2017.

Popular Stations for Round Trips

In this section I examined the most popular Bixi stations for trips that start and end at the same station. The data exploration showed that the station with the highest number of round trips is **Métro Jean-Drapeau** with **8,658** round trips. A search in Google Maps revealed that Metro Jean-Drapeau is located on St Helen's island and near a large public park, Parc Jean-Drapeau, as well as other attractions. Considering this information, this station might be a popular destination with many people taking bikes to get to and around the island.

Métro Jean-Drapeau (8,658) appears to be the most popular station for round trips



When looking at the popular stations for round trips and trips by Bixi Members, it appears that most round trips conjugated around **Métro Jean-Drapeau**, whereas member trips varied across different stations in Montreal and dropped around **Métro Jean-Drapeau**. While the latter might be a popular place for tourists and locals alike to go on a round trip, regular users might be those who take Bixi bikes to work or school across different areas of Montreal. Therefore, the Member trips are more spread out throughout the city.

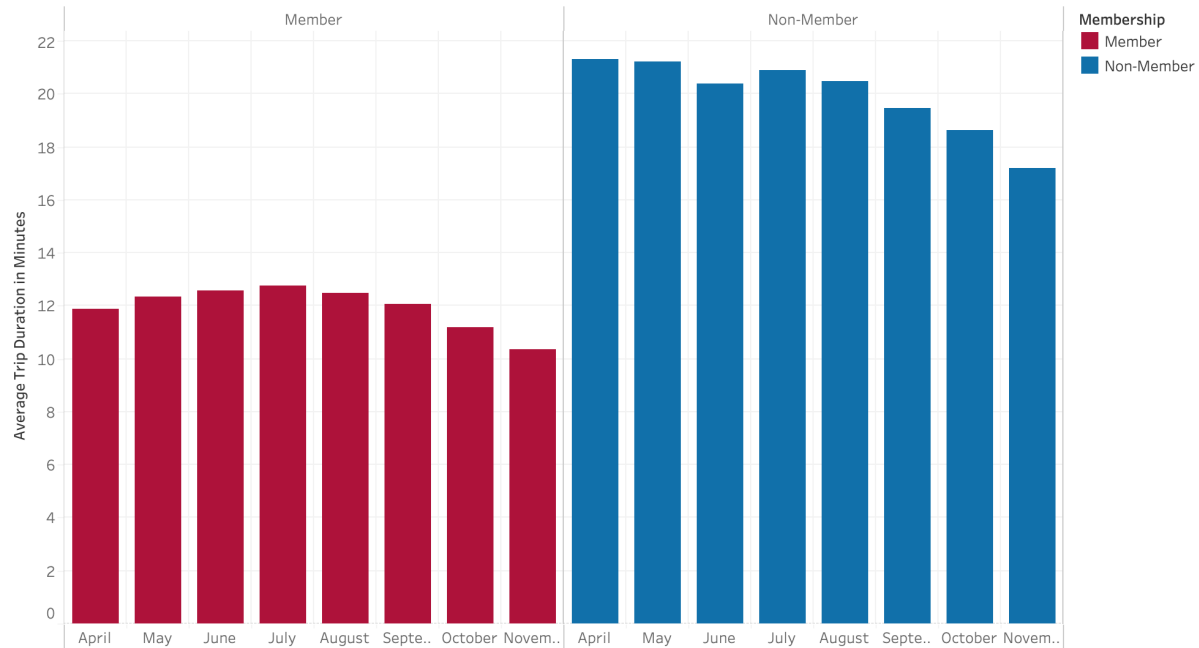
Trip Duration for Members and Non-Members

As was indicated in an earlier section of this report, Members make around 80% of all trips. Despite this, the following illustration shows that Members on average take shorter bike rides than Non-Members. The graph below illustrates that Non-Member trips tend to be almost twice

as long as Member trips. This might be due to Members taking Bixi bikes for commuting short distances every day, while Non-Bixi-Members making longer casual trips.

Difference between Member and Non-Member Trip Time in Minutes

Member average trip duration appears to be shorter than Non-Member trip duration

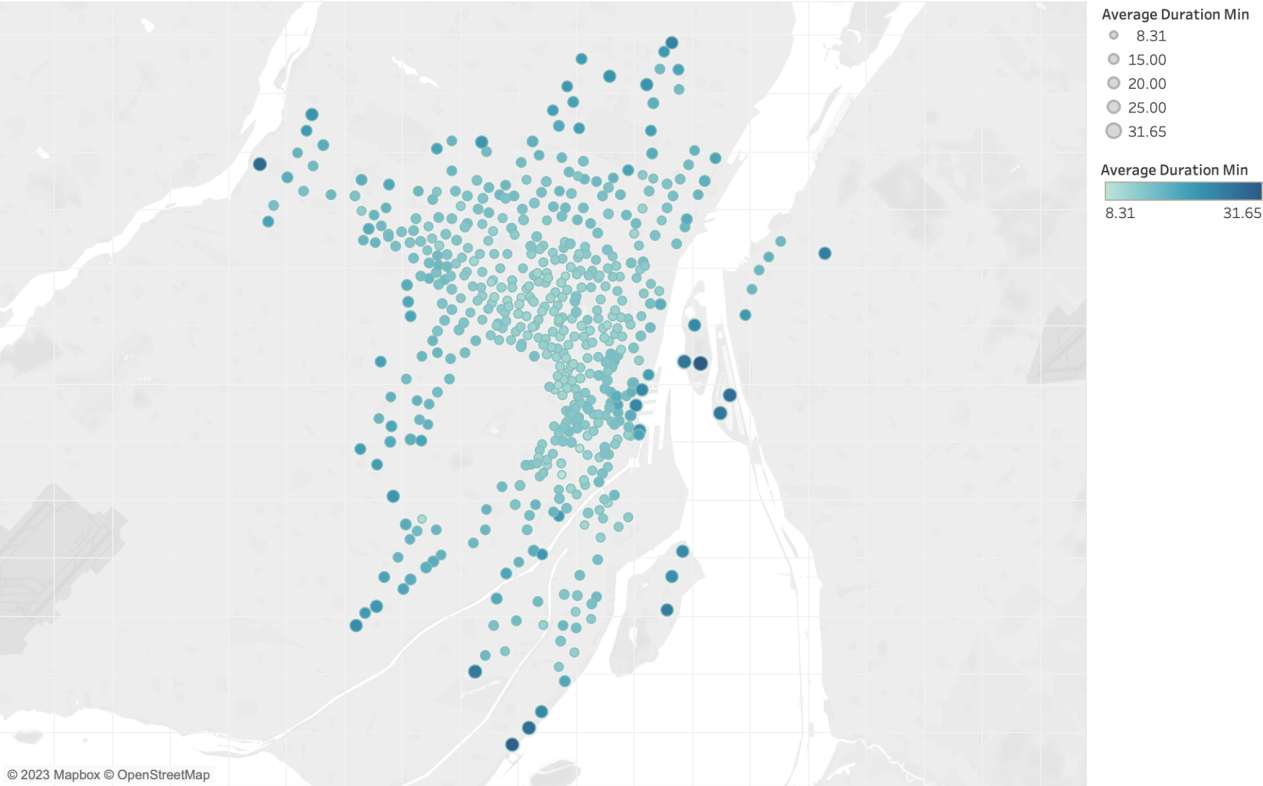


Average of Duration Min for each Start Date Month broken down by Membership. Color shows details about Membership.

Average Duration of Trips

The following two maps of Montreal illustrate the average trip duration per station. The first map shows all Bixi stations in Montreal, while the second looks closer at 5 top stations with lengthiest trips. The stations with longest trips are marked darker in color. The distribution of popular trips could indicate that the most popular stations are those located at St. Helen's island and near the Casino, both of which might be popular destinations given the attractions in those areas. Other popular stations like LaSalle/4th Avenue are located between two canals and bike paths alongside which some bike-riders might choose to explore, making the station more popular.

Montreal, QC: Average Trip Duration per Station



Map based on average of Longitude and average of Latitude. Color shows Average Duration Min. Size shows Average Duration Min. Details are shown for Station Name.

Montreal, QC: Average Trip Duration per Station

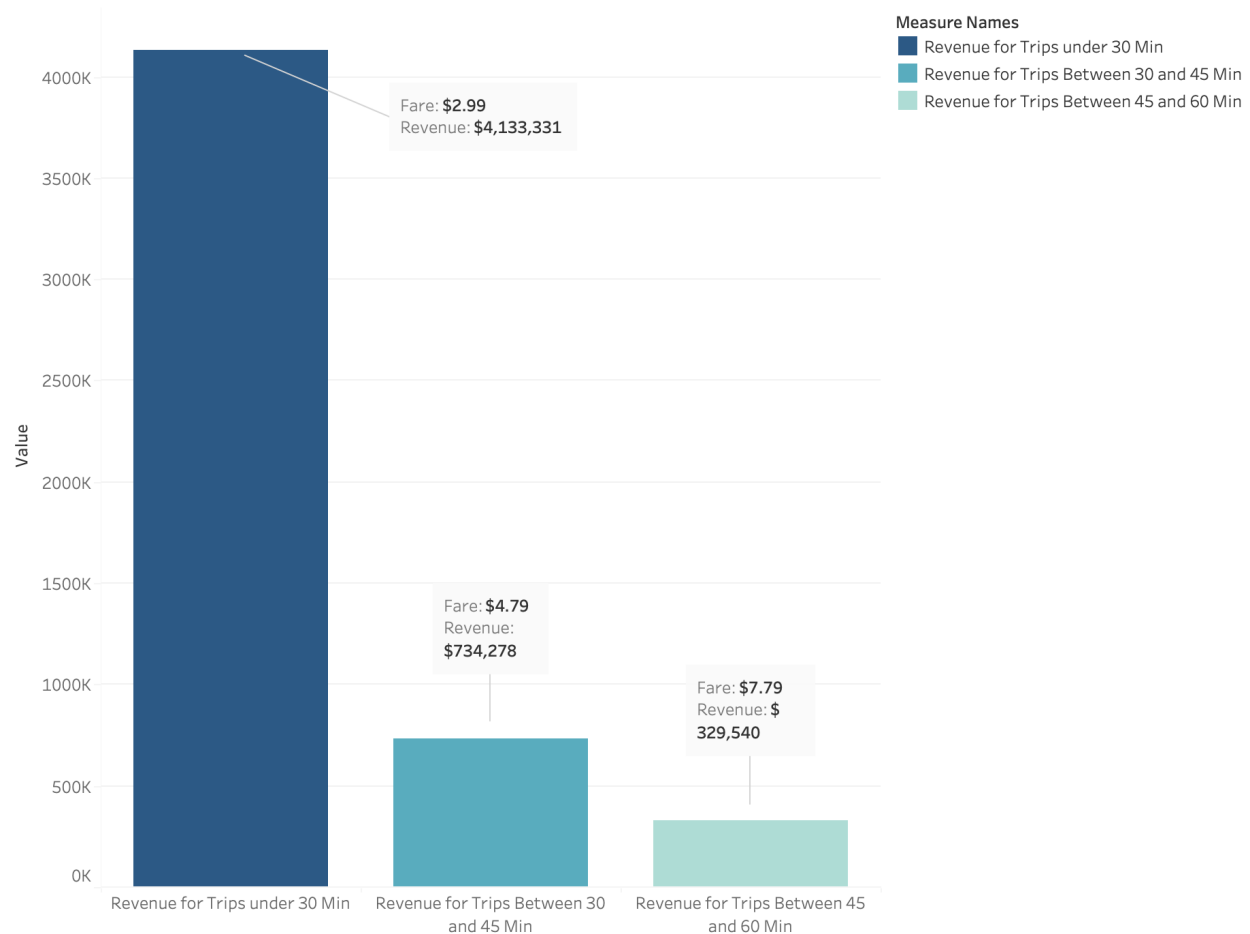


Map based on average of Longitude and average of Latitude. Color shows Average Duration Min. Size shows Average Duration Min. Details are shown for Station Name. The view is filtered on Station Name, which keeps 10 of 540 members.

Bixi's Revenue from Single Trips

In this section, I looked closer at the revenue generated by the Bixi company. Specifically, I looked at the revenue from infrequent users who make shorter trips of less than 30 minutes, trips of 30 to 45 minutes, and trips of 45 to one hour. The following are three graphs separated by the fare attached to each of the three trip durations, \$2.99 for up to 30 minute trips, \$4.79 for 30-45 minute trips and \$7.79 for 45 to one hour trips. As demonstrated in the graph, the most revenue, **\$4,133,331**, comes from single 30 minute or less trips followed by medium and long duration trips.

Revenues per Different Length Trips



Revenue for Trips under 30 Min, Revenue for Trips Between 30 and 45 Min and Revenue for Trips Between 45 and 60 Min. Color shows details about Revenue for Trips under 30 Min, Revenue for Trips Between 30 and 45 Min and Revenue for Trips Between 45 and 60 Min. The data is filtered on Membership, which keeps Non-Member.

Lastly, I looked closely at the revenue generated from trips 30 minutes or less. The data collected and visualization show that the most revenue from short trips is collected during **Saturday** and **Sunday** afternoons with a peak at around **3:00 PM**. Total revenue for 3:00 PM on Sunday is \$80,273 and \$74,239 on Saturday. Weekdays differ in that the highest revenue is generated from trips completed around **5:00 PM** regardless of the day of the week. **Fridays** at 5:00 PM showed the highest revenue of all other times during the work week and the revenue from this time period is \$45,768. In summary, Sunday afternoons generate the highest value of all.

Conclusion

This report was ordered by the executive team at Bixi to help the company derive insights from the data and grow their business. This report derived several conclusions. First, there was an overall increase in Bixi use from 2016 to 2017. Across two years there are similar patterns of use, with more trips being completed during summer and fewer in fall and spring. In terms of Bixi users, Members make the majority of trips, however, their trips are shorter on average. Meanwhile, Non-Members make less trips but their trips are longer on average. As per popular stations, Métro Jean-Drapeau appears to be the most popular station for both round trips and for the longest trips. The way Members use Bixi differs from round trips in that round trips are often completed to Métro Jean-Drapeau station while Member trips are spread out across the city. Lastly, short, 30 minutes or less trips on Saturday and Sunday afternoons tend to bring the most revenue out of all single Non-Member trips.

References

Who we are. BIXI Montréal. (n.d.). Retrieved April 16, 2023, from <https://bixi.com/en/who-we-are>